

David Harry Richman

CONTACT INFORMATION

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RESEARCH INTERESTS

Combinatorics: effective resistance, discrete potential theory, curvature on graphs
Tropical geometry: curves and Jacobians, moduli space of metric graphs
Number theory: bounds on rational points, Möbius function bounds
Phylogenetics: distance methods, metrics on tree space, applications of p -adic distance

EMPLOYMENT

National Yang Ming Chiao Tung University, Hsinchu, Taiwan
Assistant Professor, Department of Applied Mathematics 2026 – present

National Center for Theoretical Sciences, Taipei, Taiwan
Postdoctoral Fellow, Division of Mathematics 2024 – 2026

Fred Hutchinson Cancer Center, Seattle, WA
Postdoctoral Fellow, Matsen Group, Herbold Computational Biology Program 2023 – 2024

University of Washington, Seattle, WA
Postdoctoral Scholar, Department of Mathematics 2020 – 2022

EDUCATION

University of Michigan, Ann Arbor, MI
Ph.D. in Mathematics 2014 – 2020
Thesis: Weierstrass points and torsion points on tropical curves
Advisor: David Speyer

Independent University of Moscow, Moscow, Russia
Math in Moscow Study Abroad Program 2013 – 2014

Massachusetts Institute of Technology, Cambridge, MA
S.B. in Mathematics with Computer Science 2009 – 2013

RESEARCH PAPERS

“Counting two-forests and random cut size via potential theory,” with Farbod Shokrieh and Chenxi Wu, accepted to *Ann. Inst. Henri Poincaré Probab. Stat.*.

“Tropical Weierstrass points and Weierstrass weights,” with Omid Amini and Lucas Gierczak, *J. Eur. Math. Soc. JEMS* (2026), published online first.
DOI: [10.4171/jems/1792](https://doi.org/10.4171/jems/1792)

“Vector encoding of phylogenetic trees by ordered leaf attachment,” with Erick Matsen and Cheng Zhang, *Bull. Math. Biol.* **88** (2026) article 58.
DOI: [10.1007/s11538-026-01611-9](https://doi.org/10.1007/s11538-026-01611-9)

“A Ricci flow on graphs from effective resistance,” with Aleyah Dawkins, Vishal Gupta, Mark Kempton, William Linz, Jeremy Quail, and Zachary Stier, *J. Combinatorics* **16** (2025) no. 4, 553–573.
DOI: [10.4310/JOC.250923002737](https://doi.org/10.4310/JOC.250923002737)

“Node resistance curvature in Cartesian graph products,” with Aleyah Dawkins, Vishal Gupta, Mark Kempton, William Linz, Jeremy Quail, and Zachary Stier, *J. Combinatorics* **16** (2025) no. 4, 575–589.
DOI: [10.4310/JOC.250923003413](https://doi.org/10.4310/JOC.250923003413)

“The family of a -floor quotient partial orders,” with Jeffrey Lagarias, in *Proceedings of Combinatorial and*

Additive Number Theory VI (2025), 283–318.
DOI: [10.1007/978-3-031-65064-2_15](https://doi.org/10.1007/978-3-031-65064-2_15)

“Lower rational approximations and Farey staircases,” *Integers* **24** (2024), paper A37.
DOI: [10.5281/zenodo.10944039](https://doi.org/10.5281/zenodo.10944039)

“The distribution of Weierstrass points on a tropical curve,” *Selecta Math. New Ser.* **30** (2024).
DOI: [10.1007/s00029-024-00919-5](https://doi.org/10.1007/s00029-024-00919-5)

“The floor quotient partial order,” with Jeffrey C. Lagarias, *Adv. Appl. Math.* **153** (2024).
DOI: [10.1016/j.aam.2023.102615](https://doi.org/10.1016/j.aam.2023.102615)

“The tropical Manin–Mumford conjecture,” *Int. Math. Res. Not. IMRN* **2023** (2023) no. 21, 18714–18751.
DOI: [10.1093/imrn/rnad098](https://doi.org/10.1093/imrn/rnad098)

“Counting tripods on the torus,” with Jayadev S. Athreya and David Aulicino, *Arnold Math. J.* **9** (2023) 359–379.
DOI: [10.1007/s40598-022-00216-z](https://doi.org/10.1007/s40598-022-00216-z)

“Derangements and the p -adic incomplete gamma function,” with Andrew O’Desky, *Trans. Amer. Math. Soc.* **376** (2023) no. 2, 1065–1087.
DOI: [10.1090/tran/8716](https://doi.org/10.1090/tran/8716)

“Dilated floor functions with nonnegative commutators II: Negative dilations,” with Jeffrey C. Lagarias, *Acta Arithmetica* **196** (2020) no. 2, 163–186.
DOI: [10.4064/aa190628-14-1](https://doi.org/10.4064/aa190628-14-1)

“Dilated floor functions with nonnegative commutators I: Positive and mixed sign dilations,” with Jeffrey C. Lagarias, *Acta Arithmetica* **187** (2019) no. 3, 271–299.
DOI: [10.4064/aa180602-21-9](https://doi.org/10.4064/aa180602-21-9)

“Dilated floor functions that commute,” with Jeffrey C. Lagarias and Takumi Murayama, *Amer. Math. Monthly* **123** (2016) no. 10, 1033–1038.
DOI: [10.4169/amer.math.monthly.123.10.1033](https://doi.org/10.4169/amer.math.monthly.123.10.1033)

CONFERENCE
PROCEEDINGS

“PhyloVAE: unsupervised learning of phylogenetic trees via variational autoencoders,” with Tianyu Xie* (first author), Jiansi Gao, Frederick A. Matsen, and Cheng Zhang, in *Proceedings of the 13th International Conference on Learning Representations (ICLR 2025)*

PREPRINTS

“Principal minors of tree distance matrices,” with Farbod Shokrieh and Chenxi Wu, submitted.
preprint: [arXiv:2411.11488](https://arxiv.org/abs/2411.11488)

“The Möbius function on the poset of triangular numbers under divisibility,” with Rohan Pandey, submitted.
preprint: [arXiv:2402.07934](https://arxiv.org/abs/2402.07934)

“Unifying phylogenetic traversal and deep learning to guide tree exploration,” with Lena Collienne* (first author), David H. Rich, Mary Barker, Chris Jennings-Shaffer, and Frederick Matsen, submitted.
preprint: [bioRxiv:10.64898/2026.01.14.699358](https://doi.org/10.64898/2026.01.14.699358)

“On determinants of resistance matrices,” with Farbod Shokrieh and Chenxi Wu, submitted.
preprint: [arXiv:2606.21902](https://arxiv.org/abs/2606.21902)

HONORS AND
AWARDS

NSTC (Taiwan) International Conference Travel Funding for Domestic Scholars
AMS-Simons Travel Grant (\$5,000)

2025
2020 – 2022

	Rackham Predoctoral Fellowship (\$32,640)	2019 – 2020
	AMS Graduate Student Travel Grant	2019
	AMS Math in Moscow Scholarship (\$8000)	2013
TEACHING EXPERIENCE	University of Washington , Seattle, WA, USA	
	<i>Primary Instructor</i>	
	Math 208, Linear Algebra	Winter 2022, Spring 2022
	Math 308, Linear Algebra	Autumn 2020, Spring 2021
	University of Michigan , Ann Arbor, MI, USA	
	<i>Primary Instructor</i>	
	Math 116, Calculus II (Primary Instructor)	Winter 2015, Winter 2016, Winter 2018, Winter 2019
	Math 115, Calculus I (Primary Instructor)	Fall 2014
	<i>Teaching Assistant</i>	
	Math 215, Multivariable calculus (TA)	Fall 2016
	Math 216, Differential equations (TA)	Fall 2015
MENTORING EXPERIENCE	University of Washington , Seattle, WA, USA	
	WXML: Counting spanning trees on the Kagome lattice	Autumn 2021 – Winter 2022
	WXML: Zeros and critical points of complex polynomials	Autumn 2020 – Spring 2021
	University of Michigan , Ann Arbor, MI, USA	
	MREG: Rational Catalan combinatorics (online)	Summer 2021
	Laboratory of Geometry: Origami on a Hexagonal Lattice	Winter 2019
INVITED TALKS	Formalizing the matrix-tree theorem using Lean and Aristotle	
	Korean Math. Society Conf., Session on Machine-Assisted Mathematics, Seoul Korea	June 2026
	Tropical weights of Weierstrass points	
	Discrete Math Seminar, IBS DIMAG	June 2026
	Tropical Geometry in Taiwan, NCTS	August 2025
	Taipei Postdoc Seminar, NCTS	March 2025
	Informal Tropical Geometry Conference, National Taiwan University	March 2025
	AMS Fall Sectional Meeting, University of Texas San Antonio	September 2024
	(poster) WAGS 2023, University of Washington	April 2023
	Total cophenetic index and p-adic valuations	
	Workshop on Probabilistic, Combinat., and Algo. Phylogenetics, Nat'l Chengchi University	April 2026
	Planar pentagons and K3 surfaces	
	Algebraic Geometry in Xitou Workshop	July 2026
	Taiwan Math Society Annual Meeting, National Chung Cheng University	January 2026
	Tour de Formosa, Hualien to Zuoying Taiwan	January 2026
	Tree distance matrices and their minors	
	Combinatorics Seminar, University of Michigan	October 2025
	Discrete Math Seminar, University of South Carolina	October 2025
	(poster) FPSAC 2025, Sapporo Japan	July 2025
	ILAS 2025, National Sun Yat-sen University	June 2025
	National Chung Hsing University Colloquium	December 2024
	05C50 Online, University of Manitoba	May 2024

Ricci flow on graphs from effective resistance

National Taiwan Normal University Colloquium	December 2025
National Central University Colloquium	September 2025
NCTS Postdoc Symposium	September 2025
JMM 2024, Special Session on Ricci curvatures on graphs and applications	January 2024

OLA encoding of phylogenetic trees

(poster) Current Methods and Open Problems in Phylogenetics, ICERM	September 2024
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Uniform bounds for torsion packets on tropical curves

(poster) CCAAGS in honor of Bernd Sturmfels, University of Washington	June 2022
Tropical Geometry in Frankfurt (online), Goethe University Frankfurt	January 2022
Algebra and Algebraic Geometry Seminar, University of Washington	January 2021

Derangements and the p -adic incomplete gamma function

(poster) FPSAC 2022, Bangalore India	July 2022
Algebra and Algebraic Geometry Seminar, University of Washington	November 2021

Weierstrass points on tropical curves

Algebra and Number Theory Seminar, University of Kentucky	November 2019
Algebra and Algebraic Geometry Seminar, University of Washington	October 2019
SIAM Applied Algebraic Geometry, Bern Switzerland	July 2019
(poster) FPSAC 2019, Ljubljana Slovenia	July 2019
Analysis and Geometry Seminar, Central Michigan University	February 2019
Algebraic Geometry Seminar, Brown University	November 2018
Combinatorics Seminar, University of Michigan	November 2018
Algebraic Geometry Seminar, The Ohio State University	October 2018
Algebra Seminar, Georgia Tech	October 2018
(poster) AGNES Fall Meeting, Brown University	September 2018

Dilated floor functions and their commutators

AMS Fall Sectional Meeting, Madison WI	September 2019
Department of Mathematics Colloquium, University of Findlay	December 2018
INTEGERS Conference 2018, Augusta GA	October 2018
(poster) MAA MathFest, Chicago IL	July 2017
(poster) CMS Summer Meeting, University of Alberta	June 2016

SERVICE

Referee for: Journal of Integer Sequences, Electronic Journal of Combinatorics, Annals of Combinatorics, Molecular Biology and Evolution, Philosophical Transactions of the Royal Society B

Contributor to open-source software projects: SageMath, ETE (Environment for Tree Exploration) Toolkit

Co-organizer, Taipei Postdoc Seminar	Fall 2025 - Spring 2026
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Co-organizer, JMM Special Session on Ricci curvatures of graphs and applications (MRC)	January 2024
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Co-organizer, Hyperplane Arrangements Reading Group, University of Washington	Winter 2021
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Co-organizer, Student Combinatorics Seminar, University of Michigan	2018 – 2019
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Organizer, Junior Colloquium, University of Michigan	Summer 2017
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EXPOSITORY
TALKS**National Yang Ming Chiao Tung University Colloquium**

	Perspectives on tropical geometry, old and new	December 2024
	University of Findlay Colloquium (undergraduate audience) Dilated floor functions	December 2018
	Michigan Math Club (undergraduate audience) The square tile problem Descartes' rule of signs and beyond	November 2018 September 2017
	MAA MathFest , Chicago IL Descartes' rule of signs and beyond Looking for a "local" Gauss–Lucas theorem	July 2017 July 2017
	University of Washington Seminars Weierstrass points of algebraic curves and tropical curves Continuity over p -adic numbers	February 2023 November 2021
	Michigan Graduate Student Seminars What is the Jacobian of a curve? Bidding games and random-turn games Electrifying random trees II: edge correlation A brief tour of outer space Tropical Grassmannians and friends Exponentially many perfect matchings What is a tropical curve? Tate curves and Berkovich space Partition identities, generating functions, and physics What is a Néron model? Riemann–Roch on graphs How to prove the Riemann hypothesis	October 2019 March 2019 October 2018 October 2018 February 2018 October 2017 October 2016 March 2016 February 2016 January 2016 November 2015 September 2015
	Michigan Summer Mini-courses for graduate students Stratifying moduli spaces of curves by Weierstrass semigroups Combinatorial Hodge theory Tropical methods in Brill–Noether theory (5 lectures) Moduli space of tropical curves (4 lectures) Algebraic groups (5 lectures) Hodge theory for matroids (3 lectures)	Summer 2020 Summer 2019 Summer 2018 Summer 2017 Summer 2016 Summer 2016
WORKSHOPS AND CONFERENCES ATTENDED	Autoformalization for the Working Mathematician, ICERM AI for Mathematics and Theoretical CS, Simons Institute Combinatorics of Moduli of Curves, BIRS MRC on Ricci curvature on graphs and applications to data science, New York SageDays 114, IMS (Chennai, India) GATTACA Conference, Georgia Tech Arithmetic of Algebraic Curves, University of Wisconsin Tropical geometry, logarithmic geometry, and curve counting, Stockholm University Tropical geometry, mirror symmetry, and GKZ [...] KIAS (Seoul, Korea) Combinatorial Algebraic Geometry, Fields Institute Explicit Methods for Abelian Varieties, PIMS, University of Calgary Gaps between Primes and Analytic Number Theory, MSRI Arithmetic and Higher-Dimensional Varieties, University of Arizona	April 2025 April 2025 July 2024 May 2023 July 2022 March 2019 April 2018 Summer 2017 Winter 2017 Summer 2016 Summer 2016 Summer 2015 March 2015

SKILLS

- Computer: Python, Mathematica, $\text{\LaTeX}$, NetworkX, PyTorch, Scala, SageMath
- Language: English (native), Chinese (proficient), Russian (beginner)